DDGS - The Dark Horse

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Indian Livestock Sector

NOVELTECH

Livestock Industry

- 25% of Agri GDP and 4% of overall GDP
- Total LS compounded feed **>45 MMT with growth rate 5 to 8 %**
- Contributes to utilization of byproducts of Agricultural, Human Food chain, Industrial processes

Major driving factors :

- Growing population, Rising Income levels, high nutritional aspirations.
- Favourable Govt policies

Challenge:

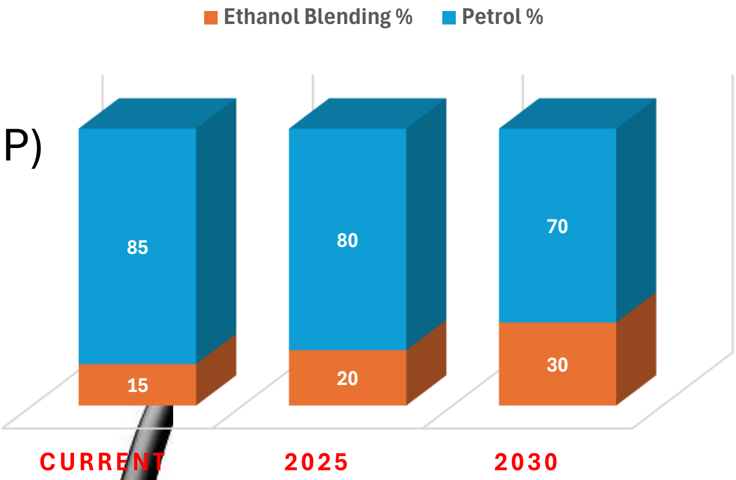
- Economic (reduce cost)
- Sustainability (Efficient Production)

Era	Year	Milk (Million Tonnes)	Eggs (Million numbers)	Wool (Million Kg)	Meat (Million Tonnes)
Post Independence	1950-51	17	1832	27.5	
Before Economic Reforms	1995-96	66.2	27187	42.4	
Basis for Future Growth	2000-01	80.6	36632	48.4	1.851
First Decade post ER	2010-11	121.8	63024	43	4.869
Second Decade post ER	2020-21	210	122050	36.9	8.798
	2021-22	221.1	129600	33	9.292
Growth in First decade post ER		51%	72%	-11%	163%
Growth up to Second decade post ER		161%	233%	-24%	375%

Source: DAHD Annual Report 2022-23

DDGS Footprint in India

The Ethanol Blending Program (EBP)



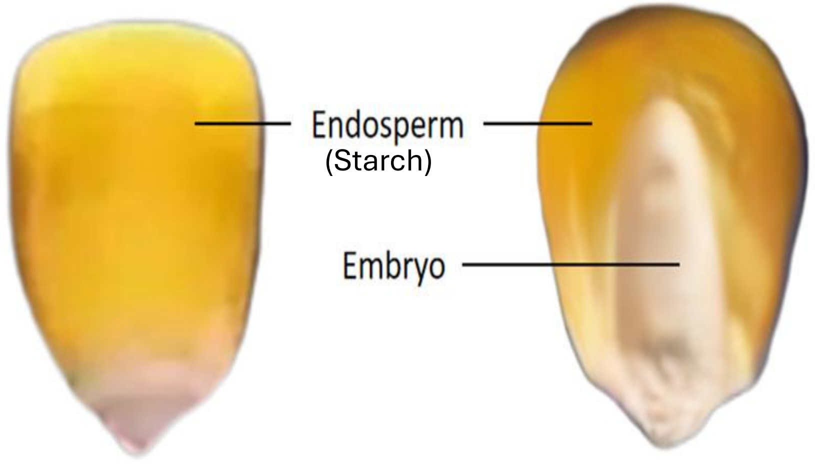
- **Since early 2000**
- Grain preferred worldwide over sugarcane - lower water and environmental footprint.
- **Stabilised Oil Industry and reduced foreign exchange outflows.**
- **Easing of strained protein scenario**

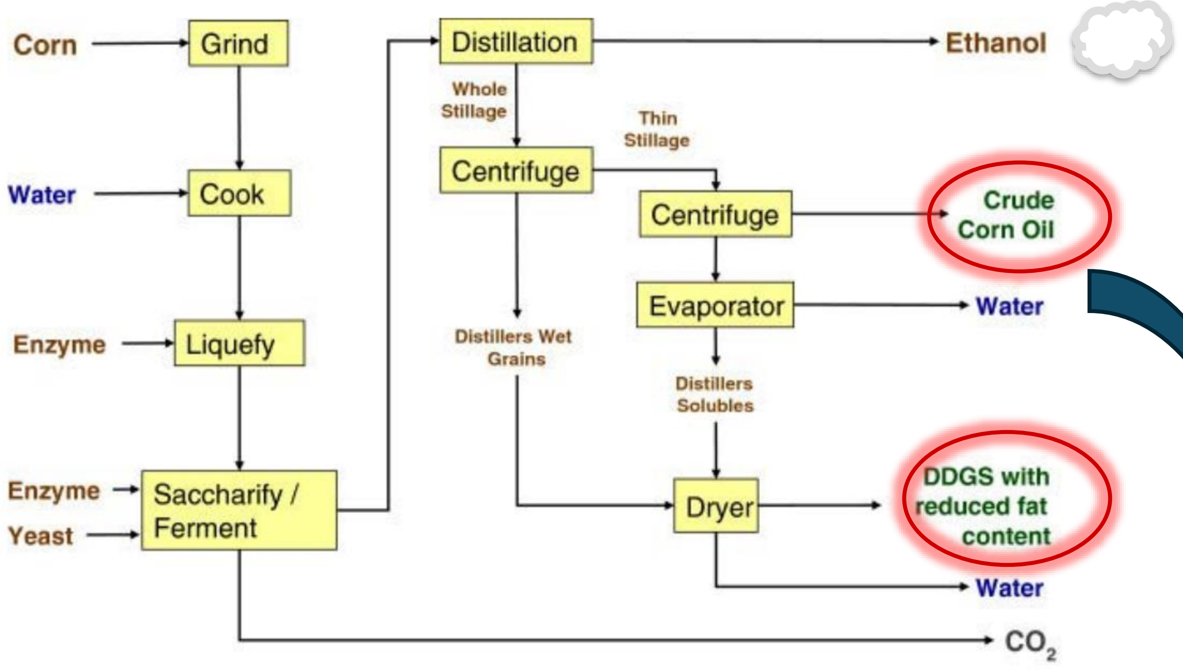
2025 ~ 500 Cr Lit Ethanol from grains

Maize requirement >60MMT (ethanol 9 MMT)

MSP @ Rs 22.5/kg + high value crop by GOI

DDGS Maize – Rice – Jowar - …..





Nutrient	Unit	Maize	DDGS Maize	
No of observations	n	15325	138	
Water	%	12.00	12.00	
Starch	**%**	**64.65**	**2.56**	
Protein	%	8.02	30.50	3.8
EE	%	4.07	11.06	2.7
NDF	%	10.62	40.22	3.8
Ash	%	1.20	4.37	3.6
Sugar	%	1.35	0.99	0.7
Non Starch Content	**%**	**25.26**	**87.14**	3.4

Rice = > 4x

Not being followed in India

What is working with DDGS?

Availability

E20 target = ~1000 Cr lit ethanol requirement = >350 Cr lit ethanol from maize

Remaining 150 Cr lit from DFG or rice?

	22-23	23-24	24-25	25-26
Grain Ethanol (Cr Lit)	124	300	400	500
DDGS MMT	0.47	1.34	1.89	2.45
Livestock Feed MMT	43	45	47	49
Penetration %	**1.1%**	**3.0%**	**4.0%**	**5.0%**

AIDA Data

Oil Marketing Companies Allocation (Figures are in Kilo Litres)

TENDER QTY	9160900	JUICE	B Heavy	C Heavy	DFG	MAIZE	TOTAL
OFFERS RECEIVED	9706833	2324865	1474349	114172	1056480	4736967	9706833
ALLOTED QTY	8368351	1887941	1139310	91555	938515	4311030	8368351
SURPLUS / NOT ALLOTED	1338482	436924	335039	22617	117965	425937	1338482
% SHARE in ALLOTED QTY		22.56%	13.61%	1.09%	11.22%	51.52%	100.00%

Source Chini Mandi

What is working with DDGS?

Nutrient	Unit	Maize	DDGS Maize	Rice	DDGS Rice	SBM
No of observations	n	15325	138	1368	792	17139
CP	%	8.02	30.50	8.22	44.84	47.31
EE	%	4.07	11.06	1.19	6.39	1.74
NDF	%	10.62	40.22	1.87	41.14	13.97
Ash	%	1.20	4.37	0.58	4.28	7.12
Sugar	%	1.35	0.99	0.31	1.70	7.26
Starch	%	64.65	2.56	77.36	2.63	0.67
Lysine	% of CP	3.11	2.67	3.54	3.16	5.95
Met + Cys	% of CP	4.39	3.83	4.93	4.31	2.65
Threonine	% of CP	3.55	3.54	3.37	3.52	3.78
Tryptophan	% of CP	0.78	0.85	1.35	1.15	1.31
Arginine	% of CP	4.82	4.63	7.77	6.53	7.28
Valine	% of CP	4.75	4.90	5.61	5.41	4.66
Isoleucine	% of CP	3.38	3.62	3.90	3.97	4.51
Leucine	% of CP	11.61	10.18	7.80	7.63	7.47

Synthetic amino acids
Toxin Binders
Enzymes
Modern formulation techniques

To whom it competes?

Soybean meal	Protein
Maize	Energy
DORB	Filler

- ✓ Since 1900
- ✓ 2/3rd of protein use
- ✓ Oil press – Heat – suppress negative factors
- ✓ Highest amino acid availability
- ✓ Defined processing
- ✓ Defined analytical procedure
- ✓ Relatively lower mycotoxin risk

- ✓ Filler - a low cost material with minimal oil.
- ✓ Does not interfere with the formulations (esp Layer)
- ✓ Comparatively controlled mycotoxins
- ✓ Good amount of starch

Before **Sourcing DDGS…**

Procurement

- Grain Sourcing – Time, place, season
- Grain moisture
- Handling
- Mycotoxin
- Grain Composition

Processing

- Dry vs Wet Processing
- Uniformity in heat treatment
- Sulphur addition
- Use of NaOH
- Fermentation regularities
- Secondary contamination

Composition?
biosafety?

Drying

- Sun drying - Old
- Machine Drying
  - Drum / Tube –
  - Vacuum
- Cooler ?

Infra in place?

Benefits and Issues: Minerals

- Sulphur:
    - H2SO4
    - Sulfur dioxide
    - Ranges between 0.4 to >1%.
- Phosphorous
    - Close to 0.6 to 1%
    - Decreased phytate
    - No threat for animal production
- Sodium
    - NaOH
    - Up to 1% detected in some samples

Safety Concerns

Environmental Pollution Concerns

Gut Health Concerns

Benefits and Issues: Protein

	Nutrient	Unit	DDGS Maize	SBM
	No of observations	n	**138**	**17139**
	CP	**%**	**30.50**	**47.31**
Poultry Specific	**Lysine**	Dig Coe %	61.00	89.00
	Met + Cys	Dig Coe %	81.00	88.00
	Threonine	Dig Coe %	69.00	83.00
	Tryptophan	Dig Coe %	81.00	89.00
	Arginine	Dig Coe %	80.00	91.00
	Valine	Dig Coe %	75.00	86.00
	Isoleucine	Dig Coe %	77.00	87.00
	Leucine	Dig Coe %	84.00	87.00
Dairy Specific	**RDP**	%	40- 50	70-75
	RUP	%	50 - 60	25-30
	RUP Dig	%	95	99

- Relatively low cost
- Digestibility concern
- Additional Protein in diet
- Imbalanced AA?
- Denaturation
- Maillard
- Gut health?

Benefits and Issues: Fat and Fat Soluble Compounds

- High Fat content –
  - High Energy
  - Formulation worries
- Natural pigments – Lutein, Zeaxanthin
- Higher PUFA - Peroxidation risk
- Palatability
- Low Milk fat syndrome

Fiber: NSP vs NDF

NDF

- Ruminal degradation
- Historical usage WDGS
- Artificial NDF/Lignin…?

NSP

- ~90 % is insoluble
- Effect on overall digestibility
- Heat treatment impact
- Use of enzymes

NDF = NSP ?

Variation = Variation in ME for poultry

Analysis by NIR?

Commercial feed mills solutions?

Mycotoxin, Antibiotic residue (?)

India, Bangladesh, Nepal

Corn Kernels (May 2023 – Apr 2024)

Parameter	Afla	ZEN	DON	T2	FUM	OTA
Number of samples	146	146	146	146	146	146
% Contaminated samples	57%	5%	11%	60%	68%	18%
Average of positives (ppb)	45	105	330	26	2281	10
Maximum (ppb)	253	643	520	64	17420	94

India, Bangladesh, Nepal

DDGS (May 2023 – Apr 2024)

Parameter	Afla	ZEN	DON	T2	FUM	OTA
Number of samples	38	38	38	38	38	38
% Contaminated samples	100%	71%	13%	82%	29%	95%
Average of positives (ppb)	42	112	2164	34	635	57
Maximum (ppb)	183	1122	9700	75	1590	441

Source: DSM

Data on Rice DDGS?
Data on Antibiotic residues?

The red flag

Cattle compatibility?

Broiler compatibility?

Colour Vs Nutrient Composition

	1	2	3	4	5
Moisture %	**10.20**	**10.80**	**11.00**	**10.20**	**10.90**
Crude Protein %	**42.10**	**47.90**	**45.50**	**45.30**	**45.80**
Crude Fat %	**4.00**	**2.70**	**4.00**	**4.60**	**4.20**
Ash %	**7.50**	**7.00**	**7.10**	**7.40**	**7.50**
Crude Fibre %	**1.20**	**1.50**	**2.50**	**1.80**	**3.60**

ADIN?

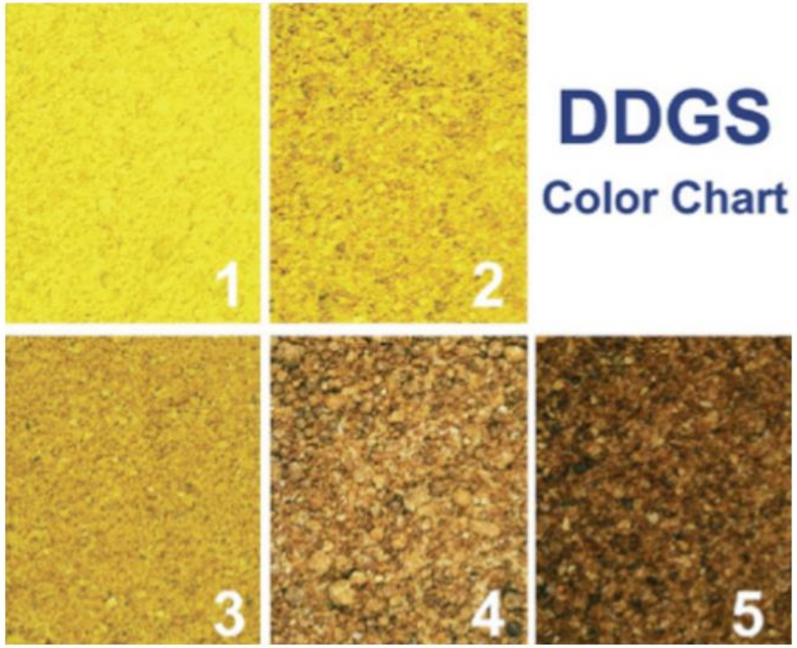
Adulteration?

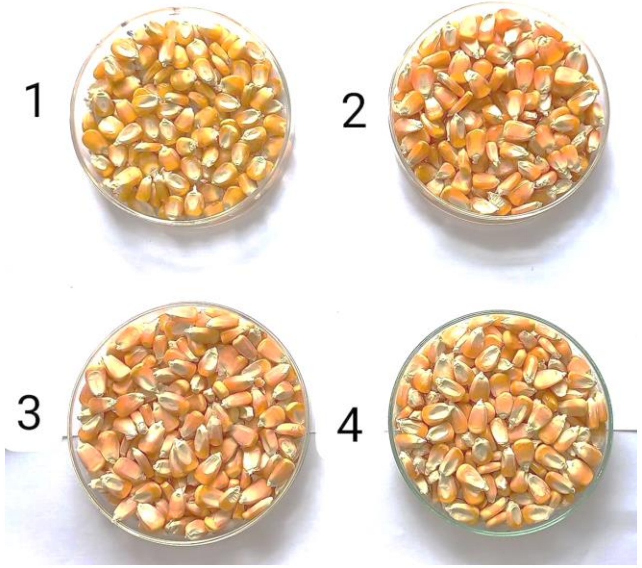
Mycotoxin?

Abc residues?

Secondary fermentation?

Colour as processing indicator?

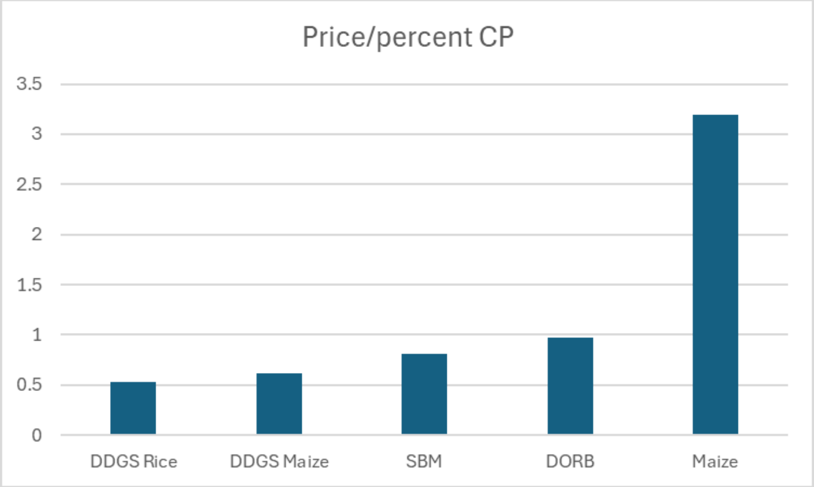


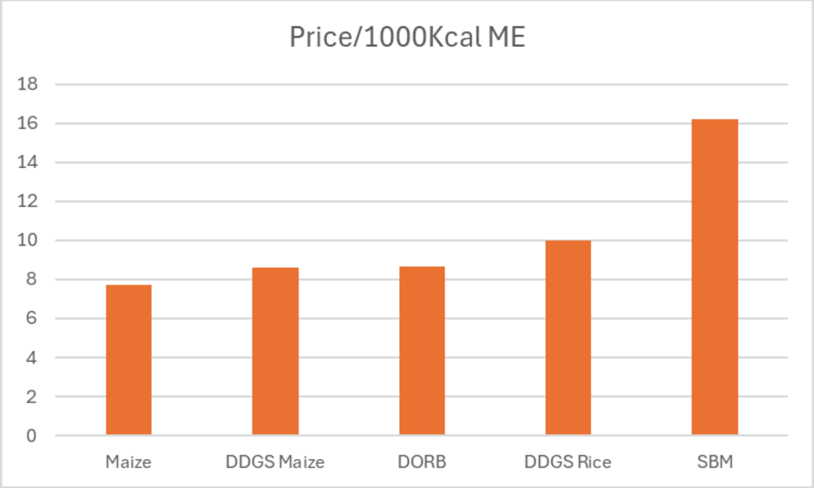


- Need to be used with caution
- Can be corelated with the smell and texture of the product
- Clear darkening of the product can be attributed to the overheating
- However, shades within yellow can not be related to the processing

	Xanthophyll c
Feed ingredient	Range*
Yellow corn	5-20
Corn gluten (60%)	150-400
DDGS**	10-50
Other ingredients	0

Price Parity





- Protein + Fat + P
- Levels of ADF and ADFIN
- Cost of
  - Enzymes,
  - Antioxidants,
  - Gut health measures,
  - Feed milling,
  - Unknown fermentation risks

Indian Raw Material Database….?

India Specific
Free Database

?

Final Comments

- **A disruptive force** in the protein scenario.
- Savings between **Rs 500 to 1000 per MT feed**.
- The recommended levels –
  - cattle feed @ **10%,**
  - Layer feed @ **5 to 7%**
  - Broiler feed @ **2 to 5%** with caution.

- **Standardisation required**
  - Plant operations
  - Specifications of products
  - Use of chemicals, additive, antibiotics etc
- Defatted DDGS.
- Analytical Data Build Up
- Low mycotoxin through genetics or processing

Thank You!